

Exp:7-Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

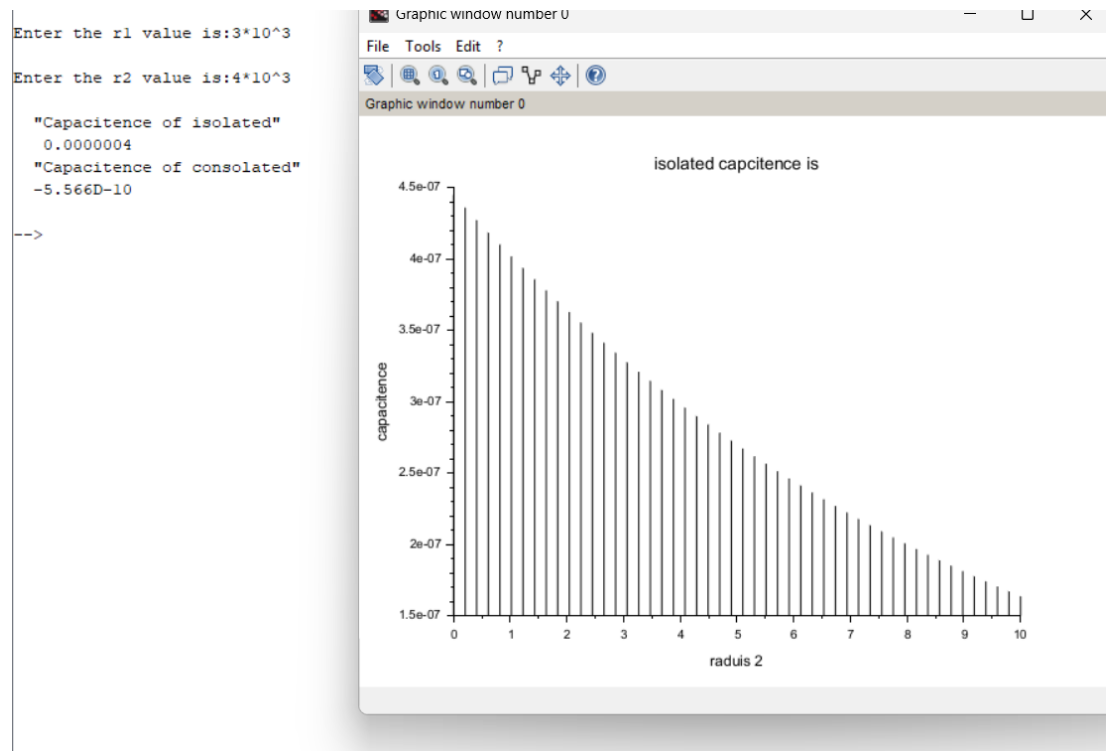
Program:

```

1 clc;
2 clear;
3 pi=3.14;
4 e=8.854*10^-12;
5
6 r1=input("Enter the r1 value is:");
7 r2=input("Enter the r2 value is:");
8 ci=4*pi*e*r2;
9 disp('Capacitance of isolated',[ci]);
10
11 r2=linspace(0,10,50);
12 x=ci*exp(-0.1*r2);
13 plot2d3(r2,x);
14 xlabel("radius-2");
15 ylabel("capacitance");
16 title("isolated capacitance is");
17 co=(4*pi*e*(r1*r2))/(r2-r1);
18 disp('Capacitance of consolated',[co]);
19

```

Output:



CALCULATION:

Exp:- 7

$r_1 = 3 \text{ mm}$

$r_2 = 4 \text{ mm}$

$\epsilon_0 = 8.854 \times 10^{-12} \text{ P/m}$

i) Isolated capacitance

$$C_{iso} = \frac{4\pi\epsilon_0 r_1 \cdot r_2}{r_2 - r_1}$$

$$C_{iso} = \frac{4\pi \times 8.854 \times 10^{-12} \times (3 \times 10^3) \times (4 \times 10^3)}{4 \times 10^3 - 3 \times 10^3}$$

$$C_{iso} = 4.0 \times 10^{-7} \text{ F} \approx 0.0000004$$

ii) Consolidated capacitance:

$$C_{con} = \frac{4\pi\epsilon_0}{\left[\frac{1}{r_1} - \frac{1}{r_2}\right]}$$

$$C_{con} = \frac{4\pi \times 8.854 \times 10^{-12}}{\left[\frac{1}{3000} - \frac{1}{4000}\right]} = 5.66 \times 10^{-10} \text{ F}$$

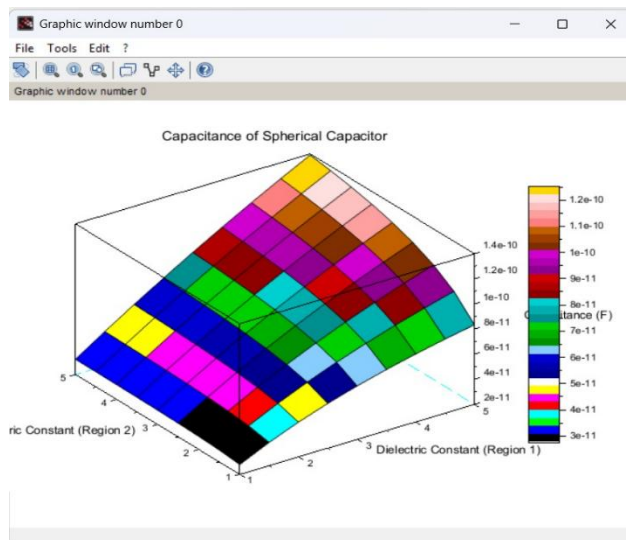
CASE 1:-CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS CODE:-

```

1 clc;
2 clear;
3 pi=3.14;
4 e=8.854*10^-12;
5
6 r1=input("Enter the r1 value is:");
7 r2=input("Enter the r2 value is:");
8 ci=4*pi*e*r2;
9 disp('Capacitance of isolated', [ci]);
10
11 r2=linspace(0, 10, 50);
12 x=ci*exp(-0.1*r2);
13 plot2d3(r2,x);
14 xlabel("radius .2");
15 ylabel("capacitance");
16 title("isolated capacitance is");
17 co=(4*pi*e*(r1*r2))/(r2-r1);
18 disp('Capacitance of consolidated', [co]);
19
20
21

```

Output:

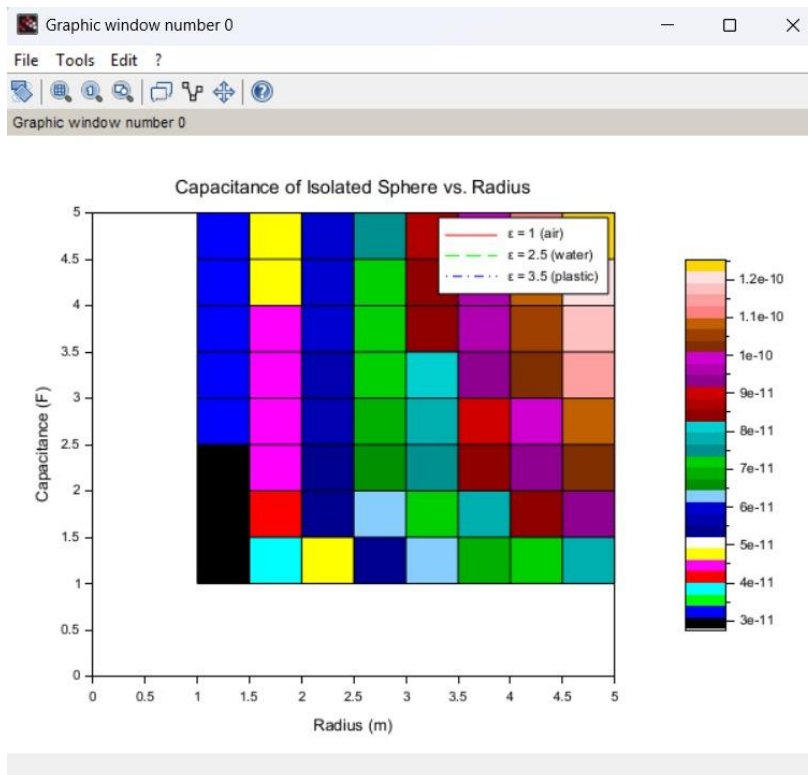


CASE 2: ISOLATED SPHERE

CODE:-

```
1 clc; clear;
2 // Constants
3 epsilon_0 = 8.854e-12;
4 R1 = 0.2; Rm = 0.6; R2 = 1.0;
5 // Dielectric constants
6 epsilon_r1 = 1:0.5:5;
7 epsilon_r2 = 1:0.5:5;
8 // Capacitance calculation
9 C = zeros(length(epsilon_r1), length(epsilon_r2));
10 for i = 1:length(epsilon_r1)
11     for j = 1:length(epsilon_r2)
12         C1 = 4 * pi * epsilon_0 * epsilon_r1(i) * (R1 * Rm) / (Rm - R1);
13         C2 = 4 * pi * epsilon_0 * epsilon_r2(j) * (Rm * R2) / (R2 - Rm);
14         C(i, j) = 1 / (1 / C1 + 1 / C2);
15     end
16 end
17 // Plotting
18 clf;
19 surf(epsilon_r1, epsilon_r2, C');
20 xlabel("Dielectric Constant (Region-1)");
21 ylabel("Dielectric Constant (Region-2)");
22 zlabel("Capacitance (F)");
23 title("Capacitance of Spherical Capacitor");
24 colorbar();
25
```

Output:



CASE 3: CONCENTRIC SPHERICAL CAPACITOR

CODE:

```
1 clc; clear; // Clear command window and variables
2 // Define parameters for isolated sphere case
3 R = linspace(0.01, 0.1, 100); // Radius from 0.01m to 0.1m
4 epsilon = [1, 2.5, 3.5]; // Different dielectric constants (e.g., air, water, and plastic)
5 epsilon0 = 8.854e-12; // Permittivity of free space (F/m)
6 C = zeros(3, length(R)); // Pre-allocate matrix for capacitance
7 // Calculate capacitance for different epsilon values
8 for i = 1:3
9     C(i, :) = 4 * pi * epsilon0 * epsilon(i) * R;
10 end
11 // Plot results
12 plot(R, C(1, :), 'r-', R, C(2, :), 'g--', R, C(3, :), 'b-.');
13 xlabel('Radius (m)');
14 ylabel('Capacitance (F)');
15 legend('ε = 1 (air)', 'ε = 2.5 (water)', 'ε = 3.5 (plastic)');
16 title('Capacitance of Isolated Sphere vs. Radius');
17 grid on;
18
```

Output:

